

# Xuedong Pan

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## PROFESSIONAL SUMMARY

Software Engineer with full-stack experience in React, C#/.NET, and modern Python web frameworks. Specialized in building 0-to-1 consumer applications, handling complex data migrations, and designing async job pipelines for heavy workloads. Quick learner who leverages AI tools to accelerate development and adapt to new tech stacks.

## PROFESSIONAL EXPERIENCE

**Software Engineer** | Sonilo Inc, Menlo Park, CA

Mar 2026 – Present

- Architected and deployed an end-to-end consumer AI music platform using .NET 8 and React, successfully scaling the infrastructure to seamlessly handle a surge of thousands of active users during launch week.
- Built an internal administrative dashboard using FastAPI (Python) and React to streamline user management, subscription tracking, and platform monitoring.
- Set up asynchronous background queues using Hangfire and Redis to handle long-running AI model inference without blocking the main web threads.
- Implemented real-time frontend updates using Server-Sent Events (SSE) for audio streaming and WebSockets for background task status.
- Designed an idempotent payment and credit system using PostgreSQL unique constraints to prevent duplicate credit assignments during Stripe webhook retries.
- Acted as the sole full-stack engineer, utilizing AI coding agents (Claude Code) to quickly ship features and deploy the containerized app to AWS ECS.

**Software Engineer** | University of Iowa, Iowa City, IA

Jun 2025 – Mar 2026

- Developed backend services for the "Cambus Ridership" transit application using .NET 8, Entity Framework, and Angular 21.
- Migrated massive historical transit datasets to SQL Server, writing custom scripts and collaborating with stakeholders to resolve legacy data inconsistencies with zero data loss.
- Built a centralized audit logging system to track sensitive data changes, making root-cause analysis significantly faster during compliance audits.
- Moved heavy CSV data export logic from the backend to the frontend using TypeScript, reducing server load and removing reliance on third-party backend libraries.

**Software Engineer Intern** | Applied Medical, Rancho Santa Margarita, CA

May 2024 – Aug 2024

- Fixed critical memory leaks in a Learning Management System by removing an unstable third-party mapping library (TinyMapper) and writing strongly-typed manual object mappers.
- Optimized data access layer for read-only APIs by bypassing ORM identity mapping, resulting in faster query execution and reduced server memory overhead.
- Improved backend test coverage from 20% to 80% by writing over 500 xUnit tests and integrating them into the Azure DevOps CI/CD pipeline.

## TECHNICAL SKILLS

Languages: Python, C#, JavaScript/TypeScript, Java, SQL, HTML/CSS

Frameworks: React, FastAPI, .NET 8, Entity Framework, Angular, Django, Flask

Infrastructure & Data: PostgreSQL, SQL Server, Redis, AWS (ECS/EC2), Docker, CI/CD

Concepts: RESTful APIs, WebSockets/SSE, Message Queues, Stripe Payment API, OAuth

## EDUCATION

**Northeastern University**, San Jose, CA

*Master of Science in Computer Science* | GPA: 4.0/4.0

Sep 2023 – May 2025

**San Jose State University**, San Jose, CA

*Bachelor of Science in Computer Science, Minor in Mathematics* | GPA: 3.99/4.0

Aug 2019 – May 2023

Honors: Summa Cum Laude, President's Scholar (2020–2022)